Can One Optical Satellite Terminal Serve 32 Users?

Beams, power budgets, tracking, and multiple access

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A satellite can already send data to one ground station at 200 Gb/s: NASA's TBIRD demonstration achieved that link rate and delivered up to 4.8 TB in a pass [1]. Now imagine a more modest aggregate demand: 32 ground terminals, each requesting 1 Gb/s. Why is this still a difficult design problem?

Each terminal needs its own message, enough received energy, and a beam that stays on target. A fast modem solves only part of this problem. Sharing the telescope can reduce duplicated hardware, but dividing power, partitioning aperture, and switching between users change what each user receives.

This tutorial follows that design choice from the optical field to delivered service. We first calculate the energy available to one user. We then compare 32 simultaneous paths with eight paths visiting four groups in turn. Finally, experiments tell us which parts of the proposed terminal are supported by evidence and which measurements determine the architecture choice.

The central result of the example: eight-way time sharing raises both active-link power and required active-link rate by a factor of four. With equal loss and no switching overhead, the received energy per delivered bit stays the same. The decision therefore turns on receiver capability, hardware cost, pointing continuity, and waiting time.

Who this is for

The assumed background is basic digital communications and linear algebra. Optical concepts are introduced when they affect a design decision. Sections 1–6 form the main reading path; the technical notes and research outlook can be read separately.

Reading path	Question
1 · Define the service	Who receives what, and when?
2 · Follow the message	What does an optical receiver actually observe?
3 · Share the physical resources	How much energy reaches each user?
4 · Compare concurrency and time sharing	What do eight paths save, and what do they cost?
5 · Read the experimental evidence	Which subsystem capabilities have been demonstrated?
6 · Make a conditional choice	Which measurements decide whether an architecture works?

1 · Define the service before choosing the terminal

Our reference scenario is **one LEO satellite sending independent downlink messages to 32 ground terminals**. We examine an interval during which all terminals are visible and acquired. Each has a 10 cm receiving aperture. A 1000 km slant range is used for all users to isolate the resource-sharing effects; an orbital simulation would replace it with a separate, time-varying range for each user.

The number 32 is a design stress case. The same reasoning applies to a general user count K. The illustrative inputs below remain fixed throughout the main comparison.

Quantity	Reference value and meaning
Service demand	1 Gb/s net payload per user; 32 Gb/s aggregate
Optical wavelength	1550 nm
Satellite aperture	20 cm diameter; total area held fixed
Laser output	4 W total while transmitting, before the listed losses
Receiving aperture	10 cm diameter per ground terminal
Beam half-divergence	10 µrad, measured at $1/e^2$ intensity
Fixed loss	6 dB for transmit optics, clear-air attenuation and receive optics; 3 dB for steering/routing
Fluctuation reserve	12 dB illustrative combined allowance for pointing and clear-air fading
Power comparison plane	Receiver optical input after the listed optical losses, before photodetection

Two service contracts, two different problems

Strict concurrency requires all 32 users to receive at once. It needs 32 independently modulated paths or another implementation providing 32 concurrent independent payload channels. **Frame-average service** requires each user to receive an average of 1 Gb/s over a declared frame. Eight paths can then visit four groups, provided active rate and delay meet the contract.

A 100 km-wide plane normal to the central line of sight subtends about 5.7° at 1000 km. This is an illustrative angular coverage requirement. Individual beams remain only microradians wide. Actual ground coverage depends on Earth projection and orbit geometry, and users' minimum angular separation still needs to be specified.

Three hardware choices will recur: replicate complete terminals; share a telescope among independent data paths; or share fewer paths over time. For a fair resource comparison, replicated terminals must also respect the same total aperture area and power. Giving every replica a separate 20 cm aperture would increase total area by 32.

The first decision is now clear: choose the service contract. Next, identify what physical resources a single independent message needs.

2 · Follow one message from bits to observations

A laser supplies an optical field. A modulator changes that field according to an electrical data waveform. A telescope shapes the outgoing beam, and steering places it on the receiver. The receiving aperture collects part of the field; the detector converts the collected light into electrical observations. Decoding recovers the message.

These functions separate two questions. The optical path determines how much light reaches a terminal. The detector determines what information the processor can extract from that light. We can calculate a received-power budget before selecting a receiver, then compare it with that receiver's sensitivity at the desired net rate and error target.

Start with a single coherent observation

A coherent receiver mixes the incoming field with a reference laser, called the local oscillator. After synchronization, a useful normalized baseband model for one received stream is:

$$y = hx + n,\ E[|x|^2] \le P.$$

Here x is a transmitted complex-field amplitude, h includes propagation and optical coupling, y is the recovered complex sample, and n describes receiver noise. The normalization makes squared field amplitude correspond to optical power. Frequency recovery, phase noise and the noise covariance remain part of the receiver model.

When several transmitted streams and observations are retained, the model becomes $\mathbf{y = Hx + n}$. Column j of H describes transmitted stream j; row i describes observation i. Several rows can belong to one user. A matrix with 32 rows therefore does not, by itself, describe 32 independent users.

Direct detection gives a different observation

A direct detector responds to intensity. With intensity modulation and direct detection (IM/DD), the input is nonnegative optical power. If contributions add incoherently in the measurement, a model of the form $y = Gp + n$ can be useful. Overlapping coherent fields instead produce intensity proportional to the squared magnitude of their sum; cross terms then matter.

Photon counting describes a detection regime: at low light levels, counts may require a Poisson likelihood with signal-dependent mean and variance. It is not a separate modulation format. Technical note A develops these models and the associated power constraints [11, 12].

For our design: first determine available receive power. Later, use a receiver sensitivity curve measured at the same optical plane, net rate and error target. The simple Gaussian rate formulas used in a few illustrations have their assumptions stated locally.

Three different meanings of “more spatial channels”

Suppose turbulence distorts a beam carrying message A. Collecting it through several apertures or optical modes gives several observations of A. Combining them can improve reception. This is **diversity**.

Now place messages A and B on different spatial field patterns and collect them at one terminal. The receiver must separate the patterns. This is **modal multiplexing**. Finally, aim A at one ground terminal and B at another. This is **multiuser beam transmission**. The physical destinations differ, even if both systems use matrix processing.

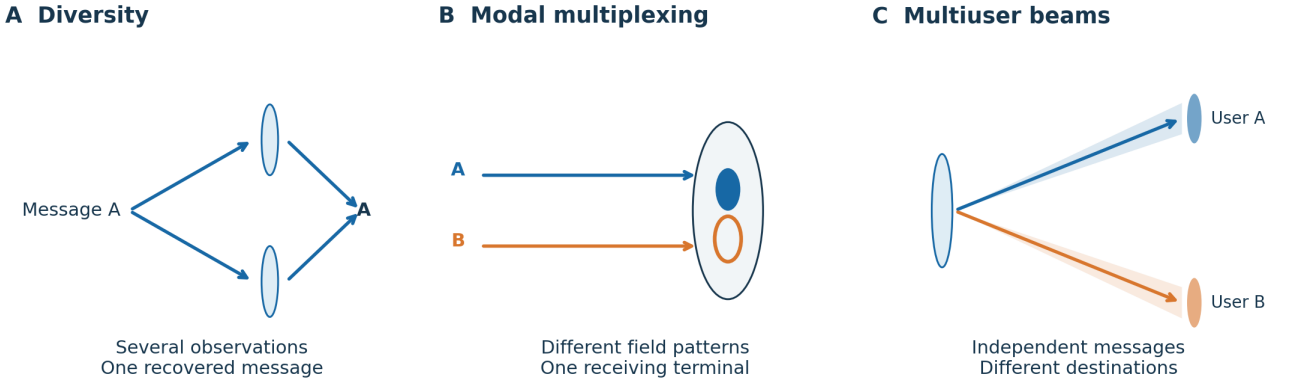


Figure 1. Three uses of spatial resources. Diversity supplies repeated observations of one message; modal multiplexing carries distinct messages to one terminal; multiuser beams address distinct terminals. Field patterns and apertures are schematic, not to scale.

Why this distinction changes the design

Our 32-user downlink primarily needs the third capability. Diversity may help a weak user, and modal multiplexing may be useful in a suitable geometry, but neither automatically supplies independent destinations. Coherently combining many emitters to form one stronger beam is another operation: those emitters may still carry only one message.

The receiving aperture also decides which field patterns survive collection. In $H = U\Sigma V^{H}$, a small singular value represents weak coupling between an input pattern and a collected output pattern. Equalization can separate sufficiently strong collected channels; it cannot replace optical energy lost outside the aperture.

For one aperture pair, a useful dimensionless geometry scale is $F = A_t A_r/(\lambda R)^2$. With our 20 cm and 10 cm apertures, $F \approx 1.0 \times 10^{-4}$. The numerator measures aperture area; the denominator brings in diffraction spreading over range. This small value motivates a weakly coupled, dominant-mode approximation for this pair. It is not a fractional mode count. Establishing several useful modes requires the singular spectrum of the actual propagation and collection operator [3].

Different ground users have different receiving apertures at different positions. Their number is not bounded by the single-pair mode count. Their useful separation depends instead on beam patterns, angular spacing, received power and pointing uncertainty.

We can now count the right thing: independent messages delivered to distinct terminals. Next, follow the power assigned to one of them.

3 · How much energy reaches one of 32 users?

With equal allocation, each simultaneous beam receives 4/32 = **0.125 W** before the listed losses. At range R = 1000 km and half-divergence θ = 10 µrad, its Gaussian $1/e^2$ intensity radius is w = Rθ = **10 m**. The receiving radius is only a = 0.05 m.

For a centered circular receiver, integrating a circular Gaussian intensity profile gives the collected fraction:

$$\eta_{geo} = 1 - \exp(-2a^2/w^2) \approx 2a^2/w^2 \approx 5.0 \times 10^{-5}.$$

The approximation uses a much smaller receiving radius than beam radius. Only about one part in twenty thousand of the beam enters the receiving aperture. This geometric loss is already large before steering loss or fluctuations are charged.

Budget step	Contribution	Remaining level
Total laser output	4 W	36.02 dBm
Divide among 32 beams	−15.05 dB	20.97 dBm
Centered geometric collection	−43.01 dB	−22.04 dBm
Fixed optical and clear-air loss	−6 dB	−28.04 dBm
Steering/routing loss	−3 dB	−31.04 dBm
Combined fluctuation reserve	−12 dB	**−43.04 dBm = 49.6 nW**

The level before the reserve is about 0.787 µW. Applying 12 dB produces a design comparison level of 49.6 nW; it does not assign that level a particular outage probability until the reserve is calibrated against measured fluctuations.

Connect aperture to the chosen beam width

For an ideal Gaussian beam, $\theta \approx \lambda/(\pi w_0)$, where w_0 is the transmit waist radius. A 10 µrad beam at 1550 nm corresponds to $w_0 \approx 49.3$ mm. A 20 cm-diameter aperture has radius 100 mm, about twice this waist. It can accommodate that illumination with little Gaussian-tail truncation in this idealized model. The aperture and divergence are therefore mutually consistent inputs; aberration, scan angle and beam quality must be checked in a real optical design.

At 1550 nm, photon energy is $hc/\lambda \approx 1.28 \times 10^{-19}$ J. The design level corresponds to about **387 arriving photons per net bit at 1 Gb/s**. Detector quantum efficiency converts arriving photons to detected events. To decide whether the link works, compare 49.6 nW with the required sensitivity at the chosen error target. Section 4 makes that comparison a function of the number of active paths.

Sharing power and partitioning aperture are separate choices

If all beams retain the same effective aperture and divergence, splitting total power among K beams costs a factor K in receive power. If the aperture area is also split into K equal pieces with the same relative Gaussian illumination, each diameter falls by √K and each divergence grows by √K. A small far-field receiver then collects another factor K less power.

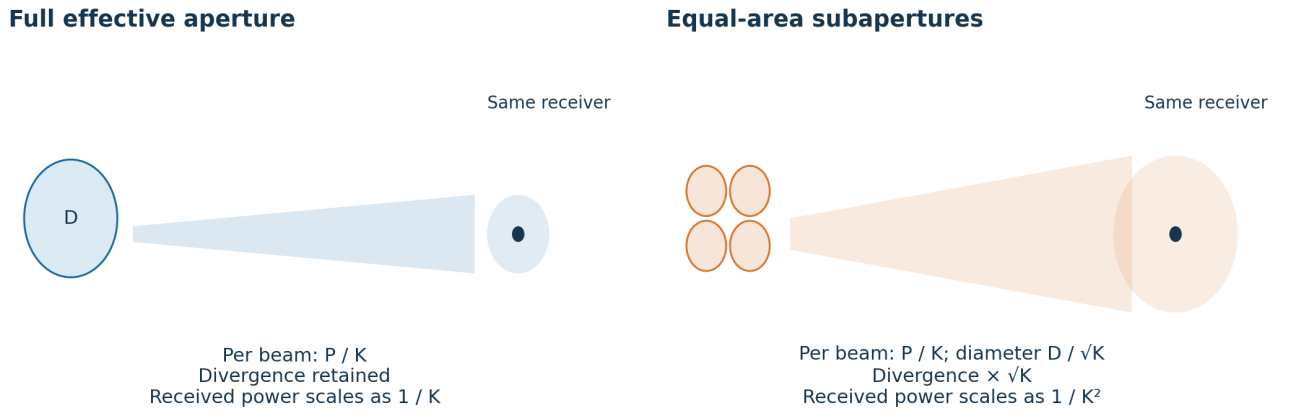


Figure 2. The two penalties have different origins. Both designs divide power; the partitioned design also broadens each spot. Four subapertures are drawn for clarity. The far-field, small-receiver approximation and equal relative illumination give 1/K versus 1/K² receive-power scaling.

In our example, dividing the 20 cm aperture into 32 equal-area pieces gives a diameter of about 3.54 cm per piece. With the same illumination ratio, divergence becomes about 56.6 µrad. The design receive level falls from 49.6 nW to approximately **1.55 nW per user**. This is why a claim of “32 beams” needs an accompanying aperture map.

What sharing the full aperture actually requires

Several angular beams can use a common pupil, but the optical network must inject independent modulated fields, route them with acceptable loss, and maintain the required beam quality across the field. Sharing a telescope alone does not guarantee that implementation. Our full-aperture budget assumes it is achieved and assigns steering/routing a provisional 3 dB loss.

The illustrative split-aperture calculation is also relevant to replicated terminals under a fixed total aperture-area budget. Replication may change pointing independence, reliability, scan loss and packaging. Those differences need their own measurements; they are not included in the geometric factor above.

The design consequence: preserve effective aperture where practical, then measure the price paid in optical efficiency and control. A saved 3 dB of real loss has the same receive-power effect as doubling total laser output. Before pursuing an even narrower beam, however, we need to examine how accurately it can be pointed.

A narrower beam helps until pointing error dominates

For a small receiver displaced by angular error r, Gaussian-beam collection has the approximate dependence $\theta^{-2}\exp(-2r^2/\theta^2)$. The factor θ^{-2} rewards concentration; the exponential penalizes missing the beam center. This already explains why maximum centered intensity need not give the best reliable service.

Assume independent, zero-mean Gaussian errors on two angular axes, each with standard deviation σ. The radial error exceeds r with probability $\exp[-r^2/(2\sigma^2)]$. To allow an exceedance probability ε, choose:

$$r_\varepsilon = \sigma\sqrt{[2\ln(1/\varepsilon)]}.$$

The receiver lies inside that angular radius with probability $1-\varepsilon$. Evaluate the received intensity at the boundary to obtain a lower power level exceeded with that probability:

$$G_\varepsilon(\theta) \propto \theta^{-2}\exp[-4\sigma^2\ln(1/\varepsilon)/\theta^2].$$

Taking the logarithm makes the optimization transparent: differentiate $-2\ln\theta - 4\sigma^2\ln(1/\varepsilon)/\theta^2$ and set the derivative to zero. The balance occurs at $\boldsymbol{\theta_{opt} = 2\sigma\sqrt{\ln(1/\varepsilon)}}$.

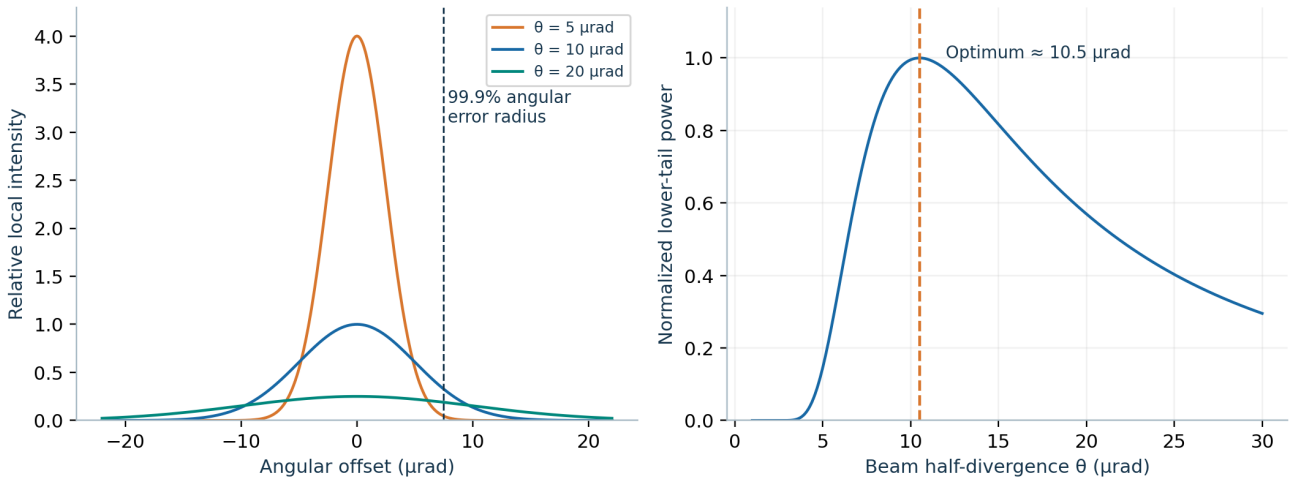


Figure 3. Left: narrowing raises centered intensity but reduces tolerance to displacement. The dashed line is the 99.9% radial-error quantile for σ = 2 μrad. Right: the resulting lower-tail power has an optimum near 10.5 μrad. These are calculated pointing-only curves, with equal transmitted power.

With σ = 2 μrad and $\varepsilon = 10^{-3}$, the optimum is about 10.5 μrad. This motivates the rounded 10 μrad used in the reference budget. At 10 μrad, the same pointing-only model gives approximately 4.8 dB loss at that quantile.

The budget's 12 dB reserve covers pointing and clear-air fading together. We therefore do not subtract that 4.8 dB again. Their joint distribution must set the eventual reserve and service availability. A diffraction limit, pointing bias, non-Gaussian tails or a finite receiver modifies the optimization; the broader pointing-outage framework is developed in [2].

Tracking quality is a communication resource

The preceding result gives prediction a direct role: reducing angular uncertainty can support a narrower beam at the same pointing-outage target. Power, aperture and estimation accuracy therefore belong in the same design comparison.

Consider the measured direction of user k as a predictable orbital component, a common satellite-attitude error, and a user-specific residual. A shared sensor can help estimate the common component. Per-user fine control handles residual errors and individual optical paths. The estimate must be propagated to the time when the beam is transmitted and arrives; sensor exposure, processing, command transfer, actuator settling and light travel all contribute.

This is a useful transfer from RF massive MIMO: use geometry and shared statistics to structure channel knowledge [10]. Its benefit is concrete here. Estimating one common attitude disturbance can assist many beams, while modeling all channel entries as unrelated can discard useful information.

More observations do not average away a shared error

If normalized branch fluctuations are $c + e_m$, with common disturbance c and mutually independent zero-mean residuals e_m independent of c, then:

$$\mathrm{Var}[(1/M)\,\Sigma_m(c + e_m)] = \mathrm{Var}(c) + \mathrm{Var}(e_m)/M.$$

Increasing M suppresses only the independent part. A shared attitude error can displace many beams together. Common cloud cover can block nearby ground terminals together. Branch correlation is therefore part of the reliability model, rather than a detail added after counting branches.

Add the atmosphere to the geometry

Turbulence changes received intensity, beam position and wavefront shape, and can reduce coupling into a selected receiver mode. A scalar irradiance distribution such as Gamma-Gamma describes a marginal power fluctuation under its assumptions [13]; it does not specify spatial-mode mixing or correlation across users. Clouds and large-scale extinction require separate availability treatment.

Our main case is downlink. Ground-to-space compensation uses a different geometry: an outgoing beam and an incoming beacon can sample different paths because of point-ahead and time evolution. Studies of guide-star/downlink measurements and downlink tomography address that inference problem [17, 18]. Their results should be transferred through the appropriate propagation model.

We now have a usable power budget and a reason to retain tracking uncertainty. The next step is to see whether sharing paths over time changes the energy requirement or mainly changes hardware and delay.

4 · Thirty-two simultaneous paths or eight shared paths?

Let M be the number of active independent paths and K = 32 the number of users. Equal time sharing gives each user duty factor M/K before overhead. To deliver average payload rate r = 1 Gb/s, the required active net rate is Kr/M. Total laser power is divided among the M active paths.

Holding divergence and the listed losses fixed, receive power also scales as 1/M. The consequence is easiest to see in a single table.

Quantity	32 simultaneous paths	8 paths, ideal sharing	8 paths, example overhead
User groups	1	4	4
Payload duty per user	1	1/4	1/4.4
Power per active beam	0.125 W	0.5 W	0.5 W
Required active net rate	1 Gb/s	4 Gb/s	4.4 Gb/s
Design receive power	49.6 nW	198.6 nW	198.6 nW
Arriving photons/net bit	387	387	352
Concurrent data paths	32	8	8
Per-user service pattern	Continuous	Periodic	1 ms per 4.4 ms

The last column uses a deliberately specified schedule: each group receives **1 ms of payload**, preceded by **0.1 ms of total switching/settling overhead**. All eight paths switch groups together. The frame lasts 4 × 1.1 = 4.4 ms. Each active path must deliver 4.4 Mbit during its 1 ms payload slot.

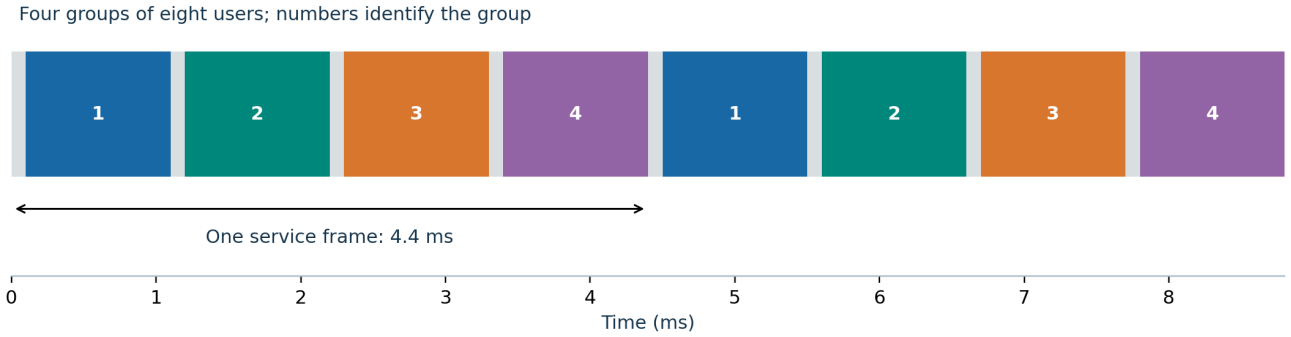


Figure 4. Two frames of the example eight-path schedule. Each colored interval serves a group of eight users; gray intervals consume 0.1 ms each. The revisit period is 4.4 ms, and the interval without payload between one user's slots is 3.4 ms. These are assumed timing inputs, not measured switching performance.

A small packet arriving just after its user's payload slot closes can wait almost 3.4 ms before the next slot, before transmission and propagation. Packet size, slot admission policy and queued traffic determine its actual delay. The ideal-sharing column leaves slot duration unspecified, so it determines duty and rate but does not determine waiting time.

Why more power per active link does not buy free bit energy

Write g for the common gain from laser output to the receiver comparison level, including the reserve used in this budget. Ignoring switching overhead:

$$P_r(M) = gP/M,\ R_{active}(M) = Kr/M.$$

$$E_{bit} = P_r/R_{active} = gP/(Kr).$$

The factor M cancels. Eight paths deliver four times the instantaneous receive power at four times the active net rate. Any receiver benefit must come from its actual rate-versus-sensitivity behavior, rather than from power per active beam alone.

With payload fraction f = 1/1.1 in the example, $R_{active} = Kr/(Mf)$, and received energy per delivered bit falls by f if laser output during active transmission remains fixed. The 4 W cap applies during both payload and overhead if the laser stays on. Blanking it during overhead saves average optical energy but does not raise payload-slot power in this comparison.

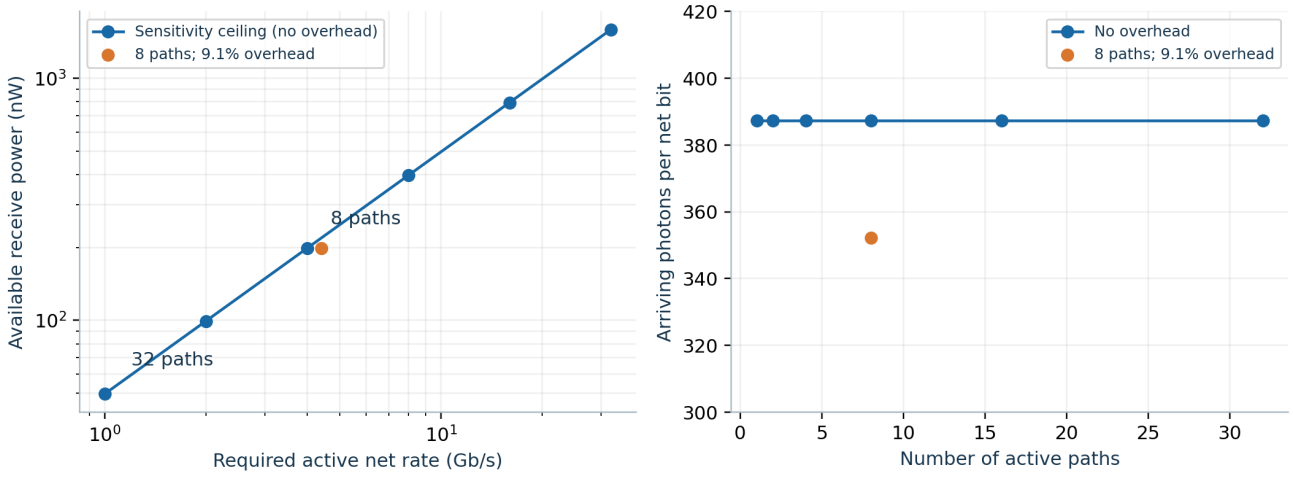


Figure 5. Left: the largest required receiver sensitivity that the example budget can support at each active net rate. A real receiver's required-power curve must lie below the relevant point. Right: ideal energy per net bit is independent of active-path count; the eight-path timing example reduces it because payload duty is lost.

For 32-way concurrency, the receiver must support 1 Gb/s at a required input power of at most 49.6 nW under the intended error and background conditions. The eight-path schedule requires at most 198.6 nW at 4.4 Gb/s. These are **sensitivity requirements**, not measured receiver specifications. They give a concrete next measurement without inventing a device curve.

Why the receiver curve can change the answer

In the coherent Gaussian reference $R = B \log_2(1 + \gamma)$, increasing rate at fixed B requires an exponential increase in SNR. Increasing B instead changes receiver bandwidth, noise and processing. At a constant spectral efficiency with noise power proportional to B, required power can approximately scale with rate. Modulation, coding, peak limits and implementation penalties decide which regime applies. Thus equal received energy per bit does not guarantee equal packet error performance.

Separate users in space first, then allocate time or wavelength

If angular separation and beam patterns provide adequate isolation, spatial-division multiple access (SDMA) lets different users reuse wavelength and modulation bandwidth. If beams overlap enough to matter, wavelength-division multiple access (WDMA) can isolate a crowded group, at the cost of extra sources, filtering and routing. Time-division multiple access (TDMA) reduces concurrent path count when the service contract permits waiting.

Observed conflict	First comparison to make	What to measure
Well-separated users, continuous demand	Concurrent spatial beams	Leakage under pointing errors; per-user power
Few data paths, delay-tolerant demand	Time sharing	Active-rate sensitivity; switching and revisit time
Angular crowding	Wavelength separation or time sharing	Filter isolation, extra loss, bandwidth and delay
One user's deep fades	Diversity or extra margin	Correlation, recovered energy and outage

A wider beam can tolerate pointing error but increase leakage. Better prediction can permit narrower beams and wavelength reuse. These interactions explain why grouping, pointing control and access allocation should use the same uncertainty model. FDMA, optical code separation and power-domain NOMA are additional options when their particular receiver costs are justified; Technical note C explains their roles.

Check service continuity for the whole group

Suppose each user's service is independently available with probability 0.99 during a declared interval. Then all 32 are available with probability $0.99^{32} \approx \mathbf{0.725}$. An all-user outage target of 1% can instead be guaranteed by allocating each user at most $0.01/32 \approx \mathbf{3.1 \times 10^{-4}}$ outage through the union bound. This sufficient allocation does not require independence, but can be conservative.

Define whether availability is conditional on visibility, clear sky and acquisition. The 12 dB illustrative reserve alone cannot establish the target. Correlated weather and common motion should be simulated jointly. Local aperture diversity can help coupling or scintillation, while geographic diversity, buffering or RF fallback may address broader outages.

Propagation adds approximately **3.34 ms one way** over 1000 km to both optical and RF links. In the scheduled example, waiting for a slot adds to that delay, together with serialization, coding, queues and possible reacquisition. Average rate is therefore only one part of the service contract.

The comparison now has three tests: sensitivity at the active rate, enough control speed to sustain duty, and acceptable per-user and joint service continuity. Experiments can tell us how credible the assumed optical and control functions are.

5 · What the experiments establish

The relevant studies address four different bottlenecks. Each example below contributes one capability to the same terminal. The experimental conditions stay attached to that capability, so the reader can see where transfer to our downlink requires an additional measurement.

Bottleneck 1 · Keep useful observations during fading

A distorted field can couple poorly into a single receiving mode even when the telescope still collects energy. One response is to collect additional modes or apertures and combine their observations of the same payload. The receiver then needs to align delays and phases while keeping the processing affordable.

Ju and colleagues demonstrated two-aperture coherent reception with a real-time adaptive equalizer [4]. For this tutorial, the useful result is that collected branches can be digitally combined in an operating receiver. Wang and colleagues studied a photonic-lantern alternative: a multimode input is mapped to several single-mode outputs, and the resulting observations are combined [9]. This moves the engineering question from “how many branches?” to “how much additional useful signal survives, at what correlation and processing cost?”

Use in our terminal: evaluate diversity on a vulnerable user at fixed total receiving area. Measure the outage reduction and compare it with spending the same resources on extra optical power. Keep branch count separate from payload count.

Bottleneck 2 · Separate messages mixed on one path

Modal multiplexing sends different messages on different field patterns. Turbulence can mix them, so the receiver must estimate and undo the mixing in the collected subspace.

Zhou and colleagues demonstrated four mode-division-multiplexed QPSK channels with a co-propagating pilot and self-coherent processing [5]. The pilot supplies information that helps separate the streams. That information has an optical-resource cost. The experiment illustrates the connection between a measurement design and a demultiplexing algorithm.

Belmonte and Kahn formulate mode selection using propagation statistics in a near-field setting [3]. This supports designing a basis for the physical channel rather than treating a named family of modes as automatically suitable.

Use in our terminal: first calculate the singular spectrum for the intended range and apertures. The weak single-pair coupling in our 1000 km example makes the number of sufficiently strong collected channels a decisive question. Modal-processing performance alone does not settle it.

Bottleneck 3 · Send independent information to distinct destinations

This is the hardware capability most directly connected to the 32-user case. Ren and colleagues report an 8 × 8 vector optical phased-array prototype, using four beams for a 100 km multipoint communication demonstration [6]. The public abstract reports a maximum 40 Gb/s communication capacity with OOK and no FEC. It does not resolve the per-beam versus aggregate accounting sufficiently for our per-user budget, so the evidence used here is the demonstrated four-beam long-range operation.

A complementary route is integrated acousto-optic steering. Lin and colleagues demonstrate independent communication using four 2 Mb/s streams at 780 nm [7]. Microwave control signals address the beams and encode their data. This is useful evidence for independent beam control and modulation in an integrated device; a link at a different wavelength and power level still needs a new budget.

The two devices should be compared at common measurement planes. Ren's reported subarray combining efficiency and Lin's waveguide-referenced steering efficiency describe different operations. End-to-end laser-output-to-user efficiency is the quantity needed to replace our assumed 3 dB steering loss.

Use in our terminal: retain independent payload paths through the shared optics, measure the effective aperture of every beam, and characterize loss over scan angle and concurrent beam count. These determine whether the 49.6 nW budget survives scaling.

Bottleneck 4 · Know where the users are when a path revisits

A switching terminal benefits if it can keep tracking users while they are not receiving payload. ESA's completed MUATS concept study addresses shared acquisition and tracking assistance, with a design for up to 20 counterpart terminals and 200 Hz angular-position updates [8]. Its breadboard uses representative simulated targets. This is evidence for a shared sensing architecture; payload throughput is a separate requirement.

For our eight-path schedule, the key quantity is the residual error when each beam arrives at its next user. Sensor updates, target identity, prediction and actuator settling jointly determine that error. A 200 Hz sensor provides an update every 5 ms; its update rate alone cannot validate the example's assumed 0.1 ms switching overhead.

Use in our terminal: track every user's state continuously, including during inactive slots, then measure full sensor-to-payload settling and reacquisition. That measurement decides whether the 4.4 ms frame is feasible.

These four examples give a division of work: diversity preserves useful observations, modal processing separates streams, multi-beam hardware addresses users, and shared tracking maintains continuity. Their combination must now pass one common service and resource comparison.

6 · Choose an architecture by its measured thresholds

For separated users requiring strict concurrency, the first candidate is **shared coarse optics with 32 independently modulated payload paths and suitable per-user fine control**. For frame-average traffic, the eight-path candidate deserves an equal-resource comparison. Aperture partitioning is a separate axis that must be evaluated for either shared or replicated hardware.

Candidate	Numerical requirement in this example	Condition that decides its value
32 full-aperture paths	Receiver supports 1 Gb/s at ≤49.6 nW	Full pupil use and independent payloads survive routing loss; 32 electronic paths fit power and thermal limits
8 full-aperture paths, example schedule	Receiver supports 4.4 Gb/s at ≤198.6 nW	0.1 ms total overhead is realized; 3.4 ms payload gap is acceptable; eight faster paths reduce actual hardware cost
32 equal-area subaperture paths	Receiver supports 1 Gb/s at approximately ≤1.55 nW	Geometric loss is offset by another measured benefit within the same total resources

These thresholds include the same provisional 12 dB fluctuation reserve. Actual outage needs its own calibration. Different architecture efficiencies change the threshold values directly.

Charge electronic power where it is consumed

A useful satellite-side bookkeeping model is:

$$P_{DC} = P_{laser}/\eta_{wall} + M\, p_{chain}(R_{active}) + P_{control} + P_{shared}.$$

P_{laser} is time-average optical output at the laser plane; η_{wall} is electrical-to-optical efficiency. The measured p_{chain} includes the modulator/driver and non-shared per-path processing at the stated rate. $P_{control}$ covers acquisition, sensing and steering; P_{shared} covers remaining common electronics. Assign each physical component once. User-terminal power is added if the objective is total-system energy.

The relevant chain comparison is **8 p_{chain}(4.4 Gb/s) versus 32 p_{chain}(1 Gb/s)**. It is not automatically a fourfold saving. Shared tracking may save hardware while faster drivers, switching and cooling consume part of that saving.

If a measured 3 dB routing loss can be removed, the receive budget improves by 3 dB without doubling laser power. If common jitter dominates, reducing residual uncertainty may be more effective than adding receivers. The next measurement should target the term that changes the architecture choice most.

Close the evaluation loop with delivered service

Use the same user trajectories, offered traffic and atmospheric realizations for each candidate. Give every architecture its own optical transfer, observation noise, control delay and power model. A controller observes noisy data, predicts the current state, selects beams and powers, and produces decoded packets. Those packets update the queues and service metrics.

Perfect future channel knowledge would remove exactly the prediction and switching costs that matter here. Instead, record what each decision knew and when it became available. Begin with equal power and round-robin scheduling, then add prediction, adaptive grouping and queue-aware allocation to identify where any improvement comes from.

Measurement	What it closes in the example
Efficiency and beam pattern versus scan angle and beam count	Replaces the assumed full-aperture mapping and routing loss
Sensitivity versus active net rate, background and error target	Tests the 49.6 nW and 198.6 nW requirements
Residual angular-error distribution and full loop delay	Sets usable divergence and pointing contribution to outage
Settling, identity retention and reacquisition time	Tests the 0.1 ms overhead and the resulting active-rate requirement
Per-path and shared electrical power	Resolves the hardware saving after faster electronics are counted
Matched-trajectory service trials	Establishes per-user rate, delay tails, joint outage and energy per delivered bit

Return to the original question

A 32-user terminal needs enough **independent optical delivery**, not simply a modem whose aggregate rate exceeds 32 Gb/s. In the reference budget, 32 full-aperture beams leave 49.6 nW per receiver after the chosen reserve. Dividing the aperture as well reduces that to about 1.55 nW under the stated illumination model.

Eight paths raise active-link power to 198.6 nW, but also require a higher active rate: 4 Gb/s ideally or 4.4 Gb/s in the timing example. They preserve ideal received bit energy and trade concurrent electronics for faster paths and waiting. Strict concurrency selects the 32-path service pattern; delay-tolerant traffic permits a measured comparison with eight paths.

The decisive next result is therefore a **joint measurement of aperture use, end-to-end efficiency, active-rate sensitivity and switching performance**. With those quantities, the designer can predict which combinations of user rate, delay and availability the terminal can actually support. That is the practical meaning of a scalable multiuser architecture.

End of the main tutorial. The following notes provide derivations, alternative observation models, access-method notes and a research outlook for readers who want to go further.

Technical note A · Observation models and receiver constraints

Intensity observations

An additive-power IM/DD model is $y = Gp + n$, with nonnegative input vector p. Mean optical power constrains $E[\Sigma p_j]$; individual peak-power limits constrain each p_j. The gain matrix G maps transmitted powers to measured outputs under the specified detector and noise approximation [11, 12].

This sum-of-powers description requires an appropriate incoherence, filtering or averaging argument. If fields overlap coherently at a detector, the incident intensity is proportional to $|\Sigma h_j E_j|^2$. Expanding that square generates cross terms. A linear propagation operator on fields and a linear observation operator on powers are therefore different assumptions.

For counts collected over interval Δt, an idealized model is:

$$N_i \mid p \sim \text{Poisson}\{\Delta t\,[\eta_q\,\Sigma_j G_{ij} p_j/(h\nu) + b_i]\}.$$

Here G maps optical power into the detector input, η_q is quantum efficiency, $h\nu$ is photon energy, and b_i is background-plus-dark count rate. Conditional independence between count channels is a separate assumption. Dead time, saturation and afterpulsing require extensions when relevant. The mean and variance both depend on intensity, so a constant-variance Gaussian approximation can be poor at low counts.

Achievable rate and cooperative references

For independent coherent Gaussian streams, treating interference as Gaussian noise gives the familiar reference $R_k = B_k \log_2(1 + \gamma_k)$. Its SINR must include actual leakage, powers and receiver noise. Non-Gaussian residuals, finite constellations and mismatched decoding may instead require mutual information, generalized mutual information, or direct coded-link evaluation.

If all receivers could pool their observations, a Gaussian covariance reference would be:

$$C_{\text{joint}} = B \max_Q \log_2 \det(I + N^{-1/2} H Q H^H N^{-1/2}).$$

Q is an admissible input covariance and N the positive-definite noise covariance. The constraint set must respect available power and hardware. Separate downlink users normally cannot pool raw observations, so this cooperative reference can bound a model without describing the actual terminal's achievable sum rate. Intensity and peak constraints need their own treatment.

Two further limits, for different questions

For an ideal pure-loss bosonic channel, a single mode with transmissivity τ and mean input photon number $\bar{n}$ has capacity $g(\tau\bar{n})$, where $g(x) = (x + 1)\log_2(x + 1) - x \log_2 x$ [14]. This is a physical information bound with optimal measurement. A terminal rate also needs the number of usable modes per second.

Finite-blocklength theory connects block size, coding rate and error probability [15]. It becomes relevant when the service contract limits coding delay. It complements propagation, switching and queueing calculations rather than determining them.

Technical note B · What the simplified rate comparison measures

Suppose K independent coherent Gaussian links each have bandwidth B, with fixed noise power and no interference. Let ρ be the SNR of one full-aperture link using all the optical power. The receive-power scalings in Section 3 give:

$$R_{full}/B = K \log_2(1 + \rho/K),$$

$$R_{split}/B = K \log_2(1 + \rho/K^2).$$

Ideal TDMA uses the full power and aperture for one user at a time. Ignoring switching, its aggregate rate is B $\log_2(1 + \rho)$. Parallel operation reuses B spatially but employs K full-bandwidth data paths. Thus the figure compares optical sharing effects, with electrical hardware still to be charged.

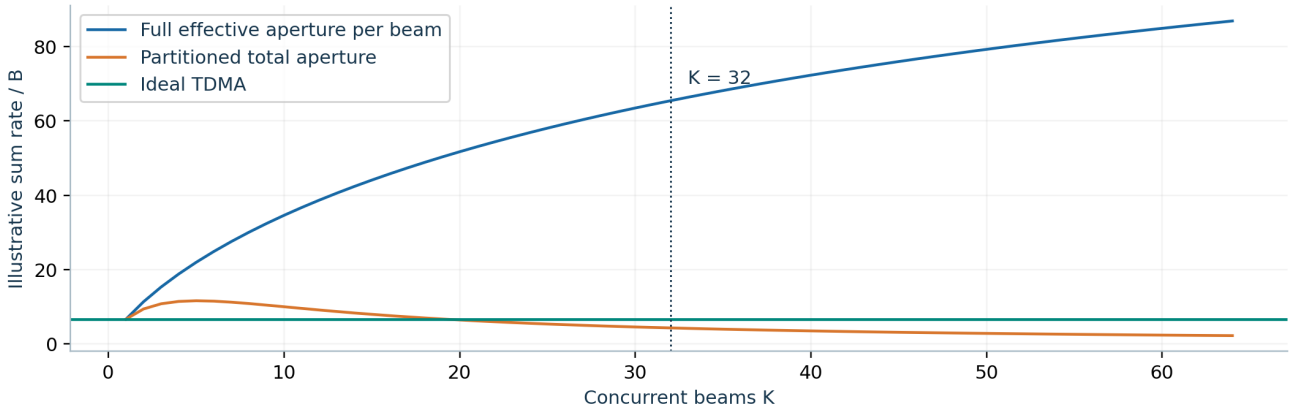


Figure 6. Calculated sum-rate illustration with ρ = 100 (20 dB). The vertical line marks 32 beams. Full-aperture sharing and aperture partitioning have very different trends, even though both divide total optical power. Each concurrent beam retains bandwidth B; electronics and device losses are excluded.

Why a fixed rate per user eventually limits K

For identical gain g and noise power σ^2, one user requiring rate r at bandwidth B needs at least $(\sigma^2/g)(2^{(r/B)} - 1)$ optical power in this Gaussian model. K independent users therefore require:

$$P_{total} \geq K(\sigma^2/g)(2^{r/B} - 1).$$

This bound assumes unchanged per-user gain and bandwidth as K grows. Aperture partitioning or extra routing loss makes g depend on K and tightens the practical constraint. The supported user count is meaningful only together with its service target and resource boundary.

Reproducing the reference calculations

The source package contains the numerical constants and figure-generation code. Geometric collection uses the exact Gaussian aperture integral; the partitioned-aperture value reported in the text uses its small-receiver scaling. The pointing optimum is obtained analytically and checked on the plotted curve. The schedule follows directly from four groups, 1 ms payload and 0.1 ms overhead per group.

The figure's Gaussian rates are separate from the receiver-agnostic optical budget. Neither the 387 photons/net bit nor the plotted Gaussian curve is a measured sensitivity claim.

Technical note C · Optional access choices and practical vocabulary

When additional access methods earn their complexity

Electrical-frequency division allocates subcarrier bands within an optical modulation/receiver chain. Wavelength division separates optical carriers and requires the corresponding optical generation, routing and filtering. Microwave tones used to control an acousto-optic steerer have yet another role. A frequency label is useful only after its physical resource is identified.

Code-division schemes require spreading codes that remain separable after the chosen intensity mapping, synchronization error and detector likelihood. Power-domain NOMA deliberately superposes users and requires a decoding order, sufficient dynamic range and a model of residual cancellation error. A four-user SLM-based experiment motivates this option [16]. Compare it with time, spatial or wavelength separation at the same power, bandwidth and service target.

A dense K-by-K equalizer with L taps scales as K^2L operations per sample in a straightforward implementation. Angular or wavelength grouping may permit smaller processing blocks. This operation count does not by itself specify electrical power; hardware architecture, precision, memory movement and sample rate also matter.

Rate and unit bookkeeping

One baud means one symbol per second. For independent, equivalently configured streams, a useful accounting relation is $R_{net} = N_{streams} R_{sym} \log_2(M_{const}) r_{FEC} f_{payload}$. The factors must describe distinct resources and overheads. If a measured input rate is already net of FEC, do not apply the code rate again. Duty, switching and retransmissions then determine delivered service over the declared interval.

Watts measure optical power; dBm is absolute power relative to 1 mW; dB is a ratio. A reference plane is the physical location at which power is specified, such as laser output, steering-network output or receiver input. Comparing efficiencies requires matching both input and output planes.

Term	Meaning used here
Aperture / spatial mode	A physical collecting or emitting opening / a field pattern used to describe transmission or reception
Field of regard / divergence	Accessible pointing directions / width of an individual beam under a stated convention
PAT / adaptive optics	Pointing, acquisition and tracking / correction of wavefront distortion
Photonic lantern	An interface mapping a multimode optical input into single-mode branches
Identity continuity	Retaining the association between each detected target and its user
Feasible service region	Rate, delay and availability combinations supported simultaneously by the architecture

Directional security. Narrow beams restrict where substantial signal energy travels. A security analysis must still specify potential interception positions, apertures, leakage and channel correlation. Authentication and encryption remain separate system functions. This topic does not alter the optical-budget comparison developed here.

Research outlook · Three experiments that would change the decision

The strongest research program follows the quantities that can reverse the architecture choice. Each task below connects a signal-processing contribution to a measurable service outcome.

Question	Concrete experiment or model	Decision-changing output
Does sharing preserve usable optical power?	Measure beam pattern, leakage and end-to-end efficiency versus scan angle and concurrent beam count; compare full and partitioned pupil use	A calibrated gain and interference model that replaces the assumed losses
Can shared inference support narrow moving beams?	Estimate common attitude and per-user residuals; predict across measured sensor and actuator delay; compare with independent tracking	Lower-tail receive power, loss-of-lock probability and reacquisition cost at fixed sensing resources
Do fewer paths actually reduce service cost?	Replay matched trajectories and traffic for 32-way concurrency and eight-path scheduling, using measured receiver curves and electronics	Delivered-rate and delay regions versus energy, path count and outage

A calibrated random propagation model specifies how chosen transmitted patterns map to collected observations as the platform and atmosphere vary. Its parameters should come from identifiable measurements. This connects finite-aperture optics, common jitter and modal coupling to estimation rather than assuming an arbitrary channel matrix.

For control, start with known geometry and equal-power scheduling. Add prediction, then adaptive beam width or user grouping, measuring the incremental service improvement. For scheduling, use realizable rates and actual switching costs; wavelength/beam/time allocation studies provide a useful networking connection [19]. Architecture studies can help identify implementable alternatives [20].

Validation can begin with four independent users to check payload separation and optical efficiency, then eight moving users to test identity and switching, and finally service emulation or hardware scaling toward 32. Each stage should retain equal-resource references, matched trajectories where possible, repeated trials and uncertainty intervals. Rare outage estimates need an explicit sampling or tail-model procedure.

Preparation and feedback

This revision was prepared with ChatGPT assistance for restructuring, explanatory writing, diagram code and illustrative calculations. Figures are original schematics or calculations from the stated models. Numerical budgets were recomputed; selected experimental claims were checked against public primary records. Literature results, design assumptions and calculated examples are identified separately throughout.

This is a technical tutorial, not a report of a new terminal experiment. Corrections and comments are welcome through the contact information on the author's website. Version date: 3 November 2026.

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